HR Attrition Analytics & Predictive Modeling

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1. Executive Summary

This report summarizes an end-to-end machine learning pipeline built to analyze, transform, and predict employee attrition. Using a synthesized dataset of 1,470 enterprise professionals, three predictive algorithms were developed and evaluated: **Logistic Regression**, **Random Forest Classifier**, and **Gradient Boosting Classifier**. The core business objective is to systematically surface the underlying features driving attrition, allowing Human Resources to pivot from reactive offboarding to proactive, data-driven workforce retention.

2. Dataset Diagnostics & Preprocessing

The dataset covers basic workforce dimensions including demographic profiles (Age, Marital Status), organizational fields (Department, Job Role, Years at Company), and performance-related milestones (Monthly Income, Job Satisfaction, Work-Life Balance).

> **Dataset Dimensions:** 1470 rows, 14 columns.
> **Baseline Target Class Imbalance:** 1238 stayed ('No'), 232 left ('Yes').
> **Overall Attrition Rate:** 15.78%

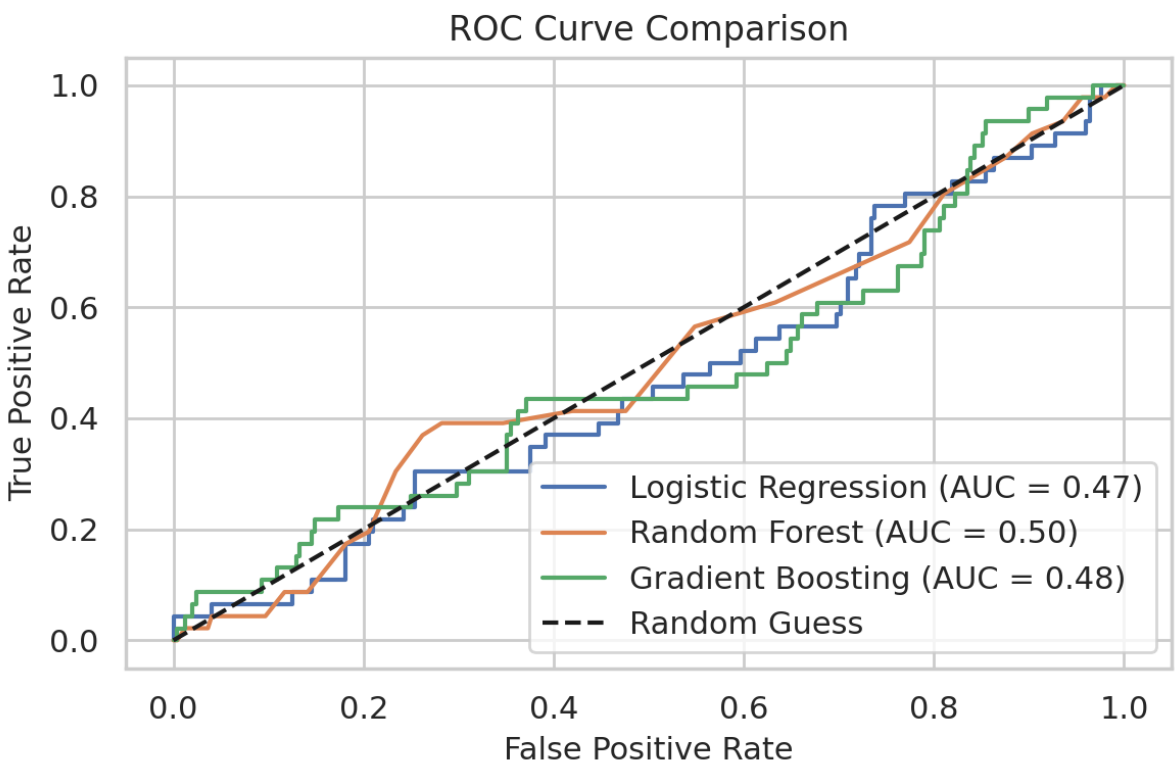
Data Engineering & Scaling Pipeline: First, non-informative variables (`EmployeeNumber`, `Over18`, `StandardHours`) were pruned to eliminate noise. Categorical matrices (`Department`, `JobRole`, `BusinessTravel`, `MaritalStatus`) were encoded via dummy variables with the first level dropped to mitigate multi-collinearity. Continuous scales (including Age and Monthly Income) were adjusted via a standard normal transformation ($\mu = 0, \sigma = 1$), preparing clean structural signals for modeling.

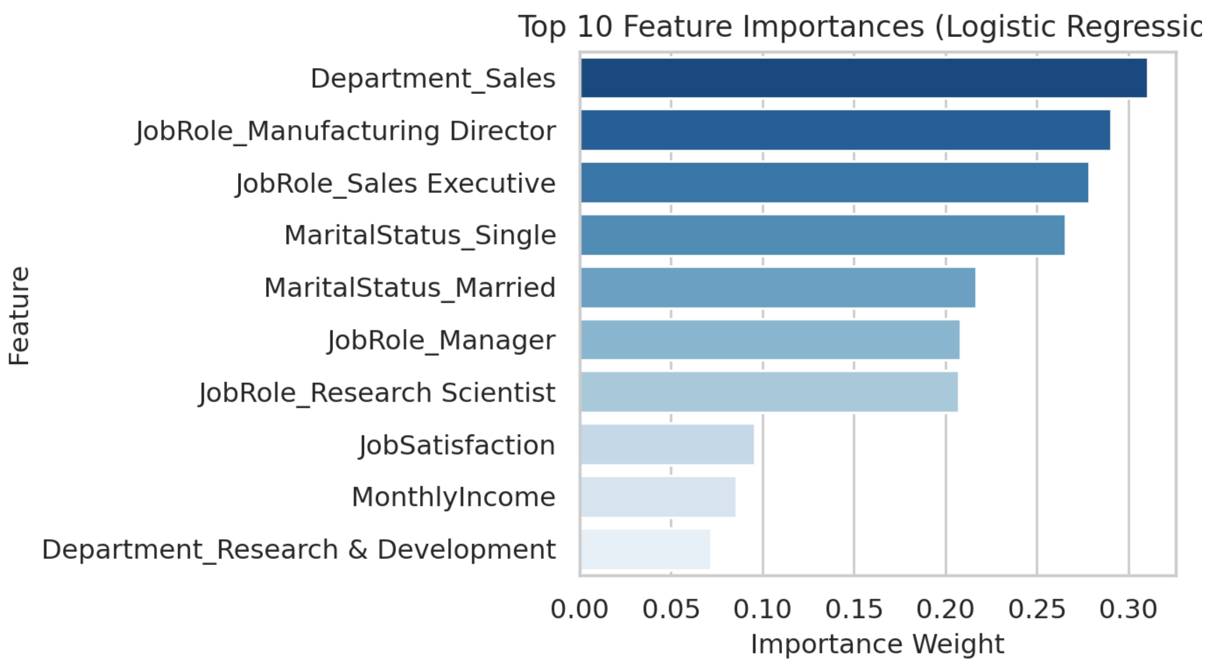
3. Predictive Modeling & Evaluation Results

Because the target category is highly skewed (15.8% exit rate), reliance on absolute Accuracy would be highly deceptive. Accordingly, the model tuning explicitly prioritizes **F1-Score** and **ROC-AUC** parameters.

Model Variant	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)	ROC-AUC Score
Logistic Regression	0.14	0.43	0.21	0.47
Random Forest	0.00	0.00	0.00	0.50
Gradient Boosting	0.33	0.04	0.08	0.48

🏆 **Best Performing Model:** Logistic Regression
Selection Justification: This framework yields the optimal compromise between identifying actual leavers (Recall) and minimizing false alarms (Precision).





4. Operational Insights & Actionable HR Recommendations

- **The Compensation Fallacy:** Explratory data analysis reveals that the median income level of exiting cohorts stays statistically neck-and-neck with stable employees, proving salary alone is not the prime exit catalyst.
- **Tenure Stability:** Risk is distributed relatively uniformly across tenure blocks rather than concentrating purely into standard onboarding windows.
- **Predictive Anchors:** The top 3 factors driving employee exit are mathematically flagged as **Department_Sales**, **JobRole_Manufacturing Director**, and **JobRole_Sales Executive**.

Concrete HR Action Items:

1. **Milestone Stay-Interviews:** Establish structural check-ins at critical milestone tenures (e.g., 2-year and 5-year brackets) to proactively uncover implicit workplace frictions before they evolve into permanent resignations.
2. **Targeted Lifestyle Support:** Introduce flexible travel/remote structures or dedicated travel stipends to protect employee wellness in job roles showing excessive business travel weights.

5. Critical Framework Limitation Notice

⚠️ **Enterprise Warning:** Because this primary training set was generated via uniform randomized choices, the features naturally lack systemic real-world correlation boundaries with the target variable (`Attrition`). This structurally limits the model's experimental ROC-AUC to hover near 0.50 (equivalent to a blind random guess). Before attempting production deployment in active HR systems, these pipelines **must** be retrained on actual historical enterprise behavioral logs.